

Worksheet-3



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Tasks

Task 1 :

1. Create a `Time` class to store hours and minutes. Implement:
 1. Overload the `+` operator to add two `Time` objects
 2. Overload the `>` operator to compare two `Time` objects
 3. Handle invalid time (`>24` hours or `>60` minutes) by throwing a custom exception

```
#include <iostream> using
```

```
namespace std;
```

```
class Time { private:
```

```
    int hours;
```

```
    int minutes;
```

```
public:
```

```
    Time(int h = 0, int m = 0) {    if (h < 0 ||  
m < 0 || h > 24 || m >= 60) {      cout <<  
"Invalid time input!" << endl;
```

```
        hours = 0;
```

```
        minutes = 0;
```

```
    } else {
```

```
        hours = h;
```

```
        minutes = m;
```

```
    }
```

```
}
```

```
    Time operator+(const Time& other) {
```

```
int totalMinutes = minutes + other.minutes;    int
totalHours = hours + other.hours + (totalMinutes / 60);
totalMinutes %= 60;

if (totalHours > 24) {    cout << "Resulting time
exceeds 24 hours!" << endl;
    return Time(0, 0);
}

return Time(totalHours, totalMinutes);
}

bool operator>(const Time& other) {    if (hours > other.hours)
return true;    if (hours == other.hours && minutes > other.minutes)
return true;
    return false;
}

void display() const {    cout << hours << " hours and " <<
minutes << " minutes" << endl;
}
};

int main() {    int h1,
m1, h2, m2;

    cout << "Enter the first time (hours and minutes): ";
```

```
cin >> h1 >> m1;

cout << "Enter the second time (hours and minutes): ";
cin >> h2 >> m2;

Time time1(h1, m1);
Time time2(h2, m2);

Time sum = time1 + time2;

cout << "The sum of the times is: ";
sum.display();

if (time1 > time2) {    cout << "The first
time is greater." << endl;
} else {
    cout << "The second time is greater or equal." << endl;
}

return 0;
}
```

Output

```
"C:\Users\aaashr\Desktop\Col  ×  +  v
Enter the first time (hours and minutes): 1 45
Enter the second time (hours and minutes): 2 35
The sum of the times is: 4 hours and 20 minutes
The second time is greater or equal.

Process returned 0 (0x0)   execution time : 7.602 s
Press any key to continue.
|
```

Task 2:

1. Create a base class `Vehicle` and two derived classes `Car` and `Bike`:
 1. Vehicle has registration number and color
 2. Car adds number of seats
 3. Bike adds engine capacity
 4. Each class should have its own method to write its details to a file
 5. Include proper inheritance and method overriding

```
#include <iostream>
```

```
#include <fstream>
```

```
using namespace std;
```

```
class Vehicle { protected:
```

```
    string RegistrationNumber;
```

```
    string Color; public:
```

```
        virtual void write() {    ofstream out("Transportation.txt",  
  
ios::app);    out << "Registration No: " <<  
  
RegistrationNumber << endl;    out << "Color: " << Color <<  
  
endl;    out.close();  
  
    }  
  
};
```

```
class Car : public Vehicle {
```

```
private:    int seats;
```

```
public:
```

```
void CarDetails(string a, string b, int seat) {  
  
    RegistrationNumber = a;  
  
    Color = b;    seats = seat;  
  
}  
  
void write() override {    ofstream  
  
file("Transportation.txt", ios::app);    file << "Car Info:\n";  
  
file << "Registration No: " << RegistrationNumber << endl;  
  
file << "Color: " << Color << endl;    file << "Seats Capacity: "  
  
<< seats << endl;  
  
    file.close();  
  
}  
  
};
```

```
class Bike : public Vehicle {  
  
private:    int engine;  
  
public:  
  
    void BikeDetails(string a, string b, int eng) {  
  
        RegistrationNumber = a;  
  
        Color = b;    engine = eng;  
  
    }  
  
    void write() override {    ofstream  
  
file("Transportation.txt", ios::app);    file << "Bike Info:\n";  
  
file << "Registration No: " << RegistrationNumber << endl;
```

```
file << "Color: " << Color << endl;    file << "Engine Capacity:
" << engine << endl;

    file.close();

}

};
```

```
int main() {

Car c1;

    Bike b1;


    string reg, colour;

int seats, engine;


    cout << "Enter Car Registration Number: ";
    cin >> reg;    cout << "Enter the Color of
the Car: ";    cin >> colour;    cout << "Enter
Number of Seats capacity: ";    cin >> seats;

c1.CarDetails(reg, colour, seats);


    cout << "Enter Bike Registration Number: ";

cin >> reg;    cout << "Enter the Colour of the
Bike: ";    cin >> colour;    cout << "Enter
```



```
Engine Capacity: ";    cin >> engine;
```

```
b1.BikeDetails(reg, colour, engine);
```

```
    c1.write();
```

```
b1.write();
```

```
cout << "Vehicle details added!" << endl;
```

```
return 0;
```

```
}
```

Output

```
"C:\Users\aaashr\Desktop\Co" × + v
Enter Car Registration Number: ba21cha4336
Enter the Color of the Car: grey
Enter Number of Seats capacity: 5
Enter Bike Registration Number: lu23cha6545
Enter the Colour of the Bike: golden
Enter Engine Capacity: 120
Vehicle details added!

Process returned 0 (0x0)   execution time : 81.310 s
Press any key to continue.
```

```
Car Info:
Registration No: ba21cha4336
Color: grey
Seats Capacity: 5
Bike Info:
Registration No: lu23cha6545
Color: golden
Engine Capacity: 120
```

2. Create a program that:

1. Reads student records (roll, name, marks) from a text file
2. Throws an exception if marks are not between 0 and 100
3. Allows adding new records with proper validation
4. Saves modified records back to file

```
#include <iostream> #include
<fstream>
using namespace std;

class error { public:
    string what() {
        return "Invalid input detected!";
    }
};

class Students { protected:
    int RollNo;
    string Name;
    float Marks;

public:
    void getinput() {
        cout << "Enter Student Details: " << endl;
        cout << "Student Roll No: ";
        cin >> RollNo;    cout <<
        "Student Name: ";    cin >>
        Name;    cout << "Marks
        obtained: ";
        cin >> Marks;

        if (Marks < 0 || Marks > 100) {
            throw error();
        }
    }

    void display() {
        cout << "Roll No: " << RollNo << endl;
        cout << "Name: " << Name << endl;    cout
        << "Marks: " << Marks << endl;
        cout << "===== " << endl;
    }
}
```

```
void readfile() {
    ifstream outFile("Student.txt");

    while (outFile >> Rollno >> Name >> Marks) {
        if (Marks < 0 || Marks > 100) {
            throw string("Invalid marks found in file!");
        }
        display();
    }

    outFile.close();
}

void writefile() {
    ofstream outFile("Student.txt", ios::app);    outFile <<
Rollno << " " << Name << " " << Marks << endl;
    outFile.close();
}
};

int main() {
    try {
        Students s1;
        s1.getinput();
        s1.writefile();
        s1.readfile();
        cout << "New record added successfully." << endl;
    }
    catch (error &e) {
        cout << e.what() << endl;
    }
    catch (string &msg) {
        cout << msg << endl;
    }

    return 0;
}
```

Output:

```
"C:\Users\ashr\Desktop\Col" × + v
Enter Student Details:
Student Roll No: 1
Student Name: Aashruti
Marks obtained: 98
Roll No: 1
Name: Aashruti
Marks: 98
=====
Roll No: 2
Name: bina
Marks: 99
=====
Roll No: 1
Name: Aashruti
Marks: 98
=====
New record added successfully.

Process returned 0 (0x0)   execution time : 38.558 s
Press any key to continue.
|
```

	File	Edit	View
1	<u>Aashruti</u>	98	
2	<u>bina</u>	99	
1	<u>Aashruti</u>	98	